Attribute-Based Searchable Encryption Scheme Supporting Efficient Range Search in Cloud Computing

Yuan Li, Haiyan Wang, Shulan Wang, and Yong Ding

***Abstract*—With the widespread application of cloud computing technology, data privacy security problem becomes more serious. The recent studies related to searchable encryption (SE) area have shown that the data owners can share their private data with efficient search function and high-strength security. However, the search method has yet to be perfected, compared with the plaintext search mechanism. In this paper, based LSSS matrix, we give a new searchable algorithm, which is suitable for many search method, such as exact search, Boolean search and range search. In order to improve the search efficiency, the 0, 1–coding theory is introduced in the process of ciphertext search. Meanwhile it is shown that multi-search mechanism can improve the efficiency of data sharing. Finally, the performance analysis is presented, which prove our scheme is secure, efficient, and human-friendly.**

***Index Terms*—multi-keyword search, multi-search mechanism, LSSS matrix, 0, 1 – coding theory, chosen-keyword attack, cloud computing**

I. INTRODUCTION

With the popularization and development of network technology and intelligent terminal [1]-[2] equipment, cloud computing technology [3]-[5] provides a huge driving force for the development of social life and economy. Especially, the integration of traditional medical system and internet technologies provides a convenient road of how to share and transfer the patients' personal health records (PHR) [6]. However, the network environment becomes more complex, which makes the traditional data sharing method face the huge challenge. Therefore, protecting the security of sensitive information is vitally important, when a growing number of medical systems share PHRs through the cloud.

Researchers are increasingly interested in searchable encryption (SE) area [7]-[11], which combines search algorithms with encryption algorithms to ensure data security and improve the efficiency of data sharing. However, there are some significant issues need to be solved. Therefore, the scheme is proposed based on CP-ABE technology [12]-[13], which is called the attribute-based searchable encryption scheme supporting efficient range search, or ABSE-ERM scheme for short. We use PHRs as an example to introduce the research background and research value of this scheme. In most telemedicine systems, patients (Data Parties) store their PHR texts (used to record patients' personal information, medical records, diagnostic data, *etc*.) in ciphertext on third-party servers, and in order to facilitate the doctor (Data Searcher) query, the patient will set multiple keywords for his PHR file, such as: "name", "gender", "age", "disease name" and "patient ID". To be noted, the "patient ID" is a unique personal identification number corresponding to the patient. When multiple patients upload their PHR ciphertext and keyword index to a third-party server, the doctor will upload their own attribute private key and search trapdoor to query the results for diagnosis. It should be noted that doctors often define a larger range for "patient ID" to obtain multiple medical records, in order to improve the diagnosis rate. Based on this circumstance, ABSE-ERM scheme provides multi-search mechanism with searchable LSSS matrix. Besides, we used a 0, 1-encoding technology to ameliorate the search efficiency, which is the first time it is used in the field of ciphertext search to deal with numeric keywords. Then we introduce the function of ABSE-ERM scheme. Firstly, we design a searchable LSSS matrix for the model according to the LSSS access policy proposed in [14]-[15], then implement a variety of search mechanisms, such as "exact search", "Boolean search" and "range search". In this paper we use the direct construction method which is different to the construction method in [16] to design the LSSS matrix. Meanwhile, we use the LSSS matrix insertion method to construct for the case of multiple threshold strings nested with each other. The scheme supports the conversion of (t, n) -threshold policy tree. When the searcher sets a larger search range, the existing CP-ABKS schemes need to establish a mark in this range. If the search range is [0,170] and the tokens are created with integers, then the schemes need to calculate 170 times for each token to complete the generation of the search token algorithm. In the search process, it is also necessary to calculate these tags 170-time to complete the matching. However, with 0, 1-coding technology, we only need to encode the upper bound of the range (assuming that the input numeric keywords are all integers greater than to 0),

This work was supported in part by the Peng Cheng Laboratory Project of Guangdong Province PCL2018KP004the National Natural Science Foundation of China under Grant 61702341, Grant 61772150.

The first two authors contributed equally to this work. H. Wang indicates corresponding author.

H. Wang is with Cyberspace Security Research Center, Peng Cheng Laboratory, Shenzhen, P. R. China (e-mail: wanghy01@pcl.ac.cn).

Y. Li is with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen, P. R. China (e-mail: 2172272145@email.szu.edu.cn).

S. Wang is with the College of Big Data and Internet, Shenzhen Technology University, Shenzhen, P. R. China (e-mail: wangshulan@sztu.edu.cn).

Y. Ding is with Guangxi Key Laboratory of Cryptography Information Security, Guilin, P.R. China, and Cyberspace Security Research Center, Peng Cheng Laboratory, Shenzhen P.R. China (e-mail: stone$_{dingy}$@126.com).

2021 IEEE Conference on Dependable and Secure Computing (DSC) | 978-1-7281-7534-8/21/$31.00 ©2021 IEEE | DOI: 10.1109/DSC49826.2021.9346237

and then calculate a few tags to complete token generation and matching. We can greatly improve the calculation efficiency and storage efficiency of the program by using the coding technology.

The main contributions of this paper are as follows:

- A searchable LSSS matrix is firstly proposed in SE area in order to implement multi-search mechanism. Besides, based on the searchable LSSS matrix, we decreased the computational and storage costs by optimizing the calculation formula.
- In order to decrease the computational costs in "range search" mechanism, the scheme in this paper applies the 0, 1-coding theory, which is firstly used to reduce the number of numeric keywords in SE area.
- Based on the proposed scheme, we give the theoretical analysis and experimental simulation to certificate the efficiency of our work.

The remaining parts of this paper are organized as follows: In Section II, related work is presented, and Section III shows some relevant cryptographic background for understanding conveniently. Then some prerequisite terms of this scheme are shown in Section IV, such as the system model and security model. Hereafter, the 0, 1–coding theory, searchable LSSS matrix, and the construction of our scheme is demonstrated in Section V. In Section VI we illustrate the performance analysis. Finally, a conclusion and further work will be provided in Section VII.

II. Related Work

How to ensure the security of the private data shared is becoming a hot research topic in the field of SE area. In 2000, the first notion of SE area was proposed by Song et al. [17], which is called symmetric SE technology (SSE) [18]-[21]. Then, a new research orientation was proposed based on public-key encryption, namely, public-key encryption with keyword search (PEKS) [22]-[25]. For example, a PEKS scheme with hidden structure was proposed by Xu et al. [26], which can improve the efficiency of keyword search, Chen et al. proposed a PEKS scheme with server aided which can resist the offline keyword guessing attack [27]. In 2019, Lu et al. proposed a certificateless keyword search encryption scheme in order to resist the outside and inside keyword guessing attacks [28]. However, there still exists an issue called the key management burden, even if the existing SE scheme can offer search service, or high-strength security.

The ciphertext-policy attribute-based encryption (CP-ABE) scheme [29]-[31] has become a practical encryption technology, which can present the provision of fine-grained access control over ciphertexts. For realized the search function, an attribute-based searchable encryption (ABSE) scheme was proposed by Wang et al. [32] in 2013, but the scheme can only provide single-keyword search. Recently, Zheng et al. [33] proposed a verifiable attribute-based keyword search scheme (VABKS) to provide multi-keyword search function. In order to improve the efficiency of search algorithm, the retrieval scheme of documents encrypted by attribute was proposed by Wang et al. in [34]. Later, The ABKS scheme with multi-keyword search was introduced in mobile crowdsourcing by Miao et al. [35].

In 2018, an efficient attribute-based multi-keyword search scheme with hybrid Boolean expression was proposed by He et al., the authors provided multi-search mechanism, and realized high security against the chosen keyword attack [36]. However, there exists a huge waste in the computational costs when the searcher uses range search mechanism, because the expression needs lots of leaf-node to be associated with a query range. The authors proposed the SE schemes based on CP-ABE technology about numeric attribute in [37] [38]. The two schemes transform the numeric attributes as strings through utilizing the 0, 1–coding theory. Unfortunately, there is no SE scheme which used the coding theory to transform the comparable numeric keyword in range search mechanism. Therefore, there are several significant advantages compared with the existing results, (*i.e.,* searchable LSSS matrix, multi-search mechanism, numeric keyword comparison, multi-keyword search, efficiency, *etc.*) in our scheme.

III. Preliminaries

A more detailed account of cryptographic background is given in the following section, which includes security assumption, and linear secret sharing scheme (LSSS). Besides, some basic parameters are introduced in Table I.

A. Generic Bilinear Group Model

Through the definition in [38] [39], let ϕ_0, ϕ_1 be the two random encodes in the additive group F_p, where $\phi_0, \phi_1 : F_p \to \{0,1\}^*, m > 3log(p)$. There is $G_0 = \{\phi_0(x)\}, G_1 = \{\phi_1(x)\}$, where $x \in F_p$. Then, the oracle computes the induced group action on G_0, G_T and a non-degenerate e; At last, the random oracle describe H_0, where G_0 is considered as a generic bilinear group.

B. LSSS matrix

Following the research in [39], each LSSS can be implemented linear reconstruction based on [40]. The detail of the definition is illustrated as follows.

Assume that the access policy $\mathbb{A}$ is a LSSS matrix, where the random attribute set belong to $\mathbb{A}$. Then, let $I \subset \{1, 2, \ldots, l\}$ be $I \subset \{i : \rho(i) \in S\}$. There exists a constant $\{\omega_I \in \mathbb{Z}_p\}$ to satisfy $\sum_{i \in I} \omega_i \lambda_i = s$, when λ_i is used to share the secure value s.

Based on [41], we can construct the corresponding LSSS matrix for (t, m)-access policy $(P_1, P_2, ..., P_n, t)$, like the formula (1).

$$M = \begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 2^2 & \cdots & 2^{t-1} \\ 1 & 3 & 3^3 & \cdots & 3^{t-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & n & n^2 & \cdots & n^{t-1} \end{pmatrix}, \rho(i) = P_i \qquad (1)$$

The secure value is shared by LSSS matrix (M, ρ) as below.

To share the secure value in (t, n)-access policy, we select a random victory $\vec{v} = (s, y_2, ..., y_t)$, where $y_2, ..., y_t \in \mathbb{Z}_p$. To any participant P_i, we afford an internal integral $\lambda_i = (M_i \cdot \vec{v})$, where M_i means the i-th row of M. Then, according to the polynomial function $f(x) = s + y_2 x + y_3 x^2 + ... + y_t x^{t-1}$, each x have the value $f(i)$. Finally, the secure value can be refactored, if and only if the acquired number of λ_i is t at least, based on the security shared scheme proposed by Shamir [42].

IV. Scheme Model

In Fig 4-1, the scheme is designed based on the cloud computing platform, and contains four entities, namely: Authorize Center (AC), Cloud Sever Provider (CSP), Data Partier (DP), and Data Searcher (DS).

1. Authorize Center: this entity runs Sea.Setup and Sea.KeyGen algorithms. AC first generates the security parameters γ, and then calculates and outputs the public key PK and the master key MSK according to the security parameters. Then, based on the attribute set of DS, AC calculates and transmits the private key SK to DS.
2. Cloud Sever Provider: this entity uses Search algorithm to return the mid-result to data searcher. It is noted that this entity is assumed semi-trust.
3. Data Partier: this entity runs Sea.Encrypt algorithm to generate ciphertext $CT_{\Lambda,W}$. To begin with, DP needs to set the access policy for each file. Then, based on PK, the files and corresponding keyword set will be encrypted, separately.
4. Data Searcher: this entity runs TrapGen and Decrypt algorithms. Firstly, DS sets several query keywords to generate the search token $Trap_{A,(M,\rho)}$ with SK in TrapGen algorithm. Secondly, when the mid-result is returned by CSP, DS runs Decrypt algorithm to decrypt it.

This paper contains six algorithms, which are defined as follows.

1. Sea.Setup(1^γ): the system setup algorithm. This algorithm inputs a security parameter for generated the public key PK and secret key MSK. PK will be delivered in public, and MSK will be storage in CA.
2. Sea.KeyGen(MSK, A): the searcher's key generation algorithm. This algorithm inputs the secret key MSK and the attribute set A to generate the private key SK for each data searcher.
3. Sea.Encrypt(PK, Λ, W, f): the searchable encryption algorithm. This algorithm inputs the public key PK, the access policy Λ, the plaintext file f and the keyword set W of the file f. Then, the ciphertext $CT_{\Lambda,W}$ will be outputted for delivered to CSP.
4. TrapGen$(SK, (M, \rho))$: the trapdoor generation algorithm. This algorithm immediately calculates the searchable trapdoor $Trap_{A,(M,\rho)}$ after received the searcher's private key.
5. Search$(CT_{\Lambda,W}, Trap_{A,(M,\rho)})$: the search algorithm. When CSP receives the trapdoor $Trap_{A,(M,\rho)}$, the algorithm will determine whether the searcher's attribute set meet the access policy Λ of ciphertext $CT_{\Lambda,W}$. If the condition is met, the keyword index of ciphertext $CT_{\Lambda,W}$ will be censored by the searchable LSSS matrix (M, ρ). If the above conditions are met, CSP will deliver the mid-result $(\tilde{C}, E)$ to the data searcher.
6. Decrypt$(\tilde{C}, E, d)$: the decryption algorithm. This algorithm decrypts the mid-result $(\tilde{C}, E)$ for restored the plaintext file f.

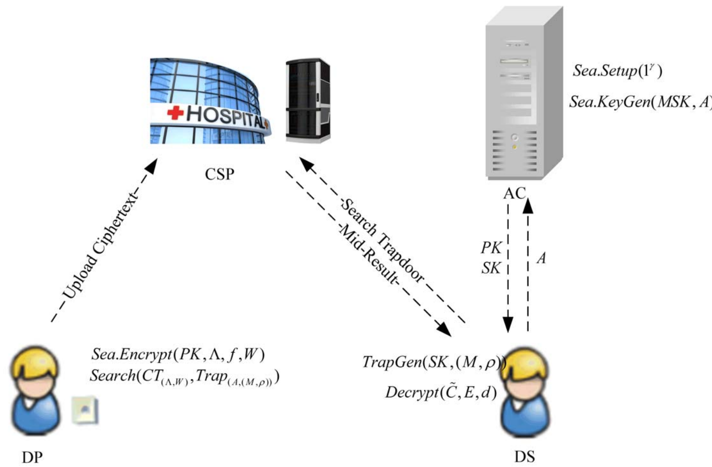

Fig. 1. The system model of ABSE-ERM scheme.

V. The Proposed ABSE-ERM Scheme

A. 0, 1 –coding theory

The coding theory [37] [38] will be used to cut down the computational costs and communicational costs of the scheme. We firstly present the detail of the coding theory. Then, an example will be given which can introduce the function of this theory.

Let $c = c_n c_{n-1} c_{n-2} ... c_1 = \{0, 1\}^n$ be a binary string of length n. It is noted that we only consider integer values in our paper. The coding rule is shown as the formula (2).

$$\begin{array}{l} X_c^0 = \{c_n c_{n-1} \dots c_{i+1} 1 \mid c_i = 0, i \in [1, n]\} \\ X_c^1 = \{c_n c_{n-1} \dots c_i \mid c_i = 1, i \in [1, n]\} \end{array} \quad (2)$$

Where the maximum number of X_c^1 and c_c^0 is $log_2 n$. We encode x as X_x^0, and encode y as X_y^1. If $X_x^0 \cap X_y^1 \neq \varnothing$, $x > y$; otherwise, $x \leq y$.

B. Searchable LSSS matrix

In this paper, based on linear secret sharing scheme (LSSS), we construct a searchable LSSS matrix structure for implemented multiple search mechanism. Besides, this structure is able to enhance the efficiency of search algorithm, compared with the Boolean keyword expression in AB-HBKS scheme. Now we elaborate the construction method of searchable LSSS matrix. We define the range of values of n is [1, 8], for any (t, n)-access structure.

Definition 1: let $A_1(P_z \to A_2)$ is a monotonous access structure over $(P_1\backslash\{P_z\}) \cup P_2$, which participant P_z is replaced with A_2 in structure A_1. For any $G \in (P_1\backslash\{P_z\})\cup P_2$, we have the equation as the follow.

$$G \in A_1\left(P_z \to A_2\right)$$
$$\Updownarrow$$
$$\{(G \cap P_1) \in A_1 \; or \; ((G \cap P_1) \cup \{P_z\} \in A_1 and G \cap P_2 \in A_2)\}$$

Based on *Definition 1*, the construction method is presented as follows. Let (M_1, ρ_1) and (M_2, ρ_2) be the access structure corresponding to the searchable LSSS matrices of A_1 and A_2, respectively. Then, the matrices are divided into the following forms.

$$M_1 = \begin{pmatrix} M_{p_z,1}, \\ \hat{M}_1 \end{pmatrix}, M_2 = \begin{pmatrix} \vec{u} & \tilde{M}_2 \end{pmatrix}$$

In this forms, assume that the sizes of M_1 and M_2 are $m_1 \times d_1$ and $m_2 \times d_2$, respectively. Let $M_{p_z,1}$ be the matrix which is composed by the row associated with participant P_z in M_1, and $\hat{M}_1$ be the matrix assembled with the rest row of M_1.

For the access structure $A_1(P_z \to A_2)$, victory $\vec{v}_i \otimes \vec{u}$ is defined as follows.

$$\vec{v}_i \otimes \vec{u} = \begin{pmatrix} u_1\vec{v}_i \\ \vdots \\ u_{m_2}\vec{v}_i \end{pmatrix}$$

In this equation, $\vec{v_i}$ means a row-victory in $M_{P_z,1} = (\vec{v}_i, ..., \vec{v}_{P_z})^T$.

Finally, matrix M is constructed as below, which is presented by the access structure $A_1(P_z \to A_2)$.

$$M = \begin{pmatrix} \vec{v}_1 \otimes \vec{u} & \tilde{M}_2 & 0 & \cdots & 0 \\ \vec{v}_2 \otimes \vec{u} & 0 & \tilde{M}_2 & \cdots & 0 \\ \vdots & 0 & 0 & \cdots & 0 \\ \vec{v}_{P_z} \otimes \vec{u} & \vdots & \vdots & \ddots & \tilde{M}_2 \\ \hat{M}_1 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

It's noted that the size of matrix M is $(m_1 + (m_2-1)p_z) \times (d_1 + (d_2-1)p_z)$.

C. The construction of ABSE-ERM scheme

ABSE-SER scheme has six algorithm: Sea.Setup, Sea.KeyGen, Sea.Encrypt, TrapGen, Search and Decrypt. Each of those algorithms will be defined as follows.

1. Sea.Setup(1_γ): In the algorithm, the authority center AC first constructs a bilinear group $e : G_0 \times G_0 \to G_T$ with a secure parameter γ, where G_0 and G_T are two cyclic groups with prime order p, and the generator g is selected. Meanwhile, two one-way hash functions $H_0 : \{0,1\}^* \to G_0$ and $H_1 : \{0,1\}^* \to \mathbb{Z}_p$ are generated. Then, AC chooses two random numbers $\alpha, \beta \in \mathbb{Z}_p$, randomly. Finally, the algorithm outputs the public key PK and master key MSK like the follows.
$$PK = (e, G_0, G_T, g, H_0, H_1, g^\alpha, g^\beta, e(g,g)^\alpha) \qquad (3)$$
$$MSK = (\alpha, \beta) \qquad (4)$$
2. Sea.KeyGen(MSK, A): The authority center AC inputs the master key MSK and the attribute set A of data searcher to create the private key SK for each searcher according to the formula (5). To begin with, AC chooses a random number $r \in \mathbb{Z}_p$ for each searcher, and then, selects a random number $r_j \in \mathbb{Z}_p$ for each attribute $j \in A$.
$$SK = \begin{pmatrix} A, D = g^{\alpha(r-\beta)} \\ \{\forall j \in A : D_j = g^r H_0(j)^{r_j}, D'_j = g^{r_j}\} \end{pmatrix} \qquad (5)$$
3. Sea.Encrypt(PK, λ, f, W): In this algorithm, the data partier DP will create the ciphertext with the public key PK, access policy λ, plaintext file f and the keyword set W of plaintext file f. The searchable encryption operation is executed as the following steps.
   1) DP selects a random number s_0 as the secure value, and then, calculates C' and $\tilde{C}$ in the formula (6), respectively.
   2) Based on the access policy, let Y be the leaf-node set of access policy λ. Then, for each leaf-node $\forall y \in Y$, DP computes C_y and C'_y with the formula (7).
   3) For each keyword $W_j = (w_1, ..., w_k), j \in [1, k]$, DP chooses a random number to calculate C_w and C'_w based on the formula (8).

   Finally, the ciphertext $CT_{\lambda,W}$ will be generated like the formula (9).
$$C' = g^{s_0}, \tilde{C} = f \cdot e(g,g)^{\alpha s_0} \qquad (6)$$
$$\forall y \in Y : C_y = g^{q_y(0)}, C'_y = H_0(att(y))^{q_y(0)} \qquad (7)$$
$$\left\{C_w = g^{\beta s_0} \cdot g^{H(w_i)s_i}, C'_w = g^{s_i}\right\}_{w_i \in W} \qquad (8)$$
$$\left(\Lambda, W, C', \tilde{C}, \left\{C_y, C'_y\right\}_{y \in Y}, \left\{C_w, C'_w\right\}_{v \in \mathbb{W}}\right) \qquad (9)$$
4. TrapGen$(SK, (M, \rho))$: DS creates the searchable LSSS matrix (M, ρ) for the query keyword set Q, where the map function is $\rho(i) = j, j \in Q$. Then, the search trapdoor is generated as the following steps after received the private key SK.
   1) DS selects two random numbers $t, d \in \mathbb{Z}_p$, and then, calculates according to the formula (10). In particular, the random number d is only stored locally.
   2) For $\forall j \in A$, DS computes $\hat{D}_j$ and $\hat{D}'_j$ as the formula (11).
   3) In order to search the ciphertext, the searchable LSSS matrix (M, ρ) is encrypted after created a query keyword set $Q = (k_1, ..., k_m)$. First, DS selects a random vector $\vec{v} = (s_0, y_2, \ldots, y_n)^T \in \mathbb{Z}_p^l$, where $y_i \in \mathbb{Z}_p$ is selected for sharing the random number t. Second, DS computes $\lambda_i = M_i \cdot \vec{v}$, where M_i represents the corresponding row of the i-th keyword in the matrix. Finally, DS calculates the formula (12).

   Then, the search trapdoor $\mathrm{Trap}_{(A,(M,\rho))}$ can be expressed as the formula (13).
$$\hat{D} = D^t \cdot g^{\alpha d} = g^{\alpha(r-\beta)t+\alpha d} \qquad (10)$$

$$\forall j \in A: \hat{D}_j = D_j^t = g^{\alpha tr} H_0(j)^{tt_j}, \hat{D}_j' = \left(D_j'\right)^t = g^{tt_j} \tag{11}$$

$$\left\{\hat{D}_k = g^{-\alpha H(k)\lambda_k}, \hat{D}_k' = g^{\alpha\lambda_k}\right\}_{k\in Q} \tag{12}$$

$$\left(A, (M,\rho), \hat{D}, \left\{\hat{D}_j, \hat{D}_j'\right\}_{j\in A}, \left\{\hat{D}_k, \hat{D}_k'\right\}_{k\in Q}\right) \tag{13}$$

5. Search$(CT_{\Lambda,W}, Trap_{A,(M,\rho)})$: After receiving the search trapdoor $Trap_{A,(M,\rho)}$, CSP executes this algorithm with the ciphertext $CT_{\Lambda,W}$. The implementation process is introduced as the following steps.
   A. CSP determines whether the attribute set A of DS meets the requirement of the access policy Λ of $CT_{\Lambda,W}$.
      a. If y is the leaf-node in Λ, let $att(y)$ be an attribute j. For each attribute $j \in A$, the mid-value E_y is calculated based on the formula (14).
      b. If y is the non-leaf node in Λ, let z be a set of child nodes of arbitrary k_z-size, and then, CSP calculates the mid-value E_y as the formula (15). If y is the root node, the mid-value E_R is output as the formula (16).

$$E_y = \frac{e\left(C_y, \hat{D}_j\right)}{e\left(C_y', \hat{D}_j'\right)} = e(g,g)^{arq_y(0)t} \tag{14}$$

$$E_y = \prod_{\varepsilon\in S_y} E_z^{\Delta_{i,S_y'},(0)} = e(g,g)^{\alpha rq_y(0)t} \tag{15}$$

$$E_R = e(g,g)^{rtq_R(0)} = e(g,g)^{arts_0} \tag{16}$$

   If the above calculation can be completed, CSP executes the following calculation. Otherwise, CSP outputs an error identifier "$\perp$".
   B. Based on the keyword set $W_i = (w_1, \ldots, w_k), i \in [1,k]$ of each ciphertext, CSP determines whether there is a ciphertext that satisfies the searchable LSSS matrix (M,ρ). The calculation method is presented as follows.
   From the definition of the LSSS secret sharing policy, we can be decrypted and n represents the total number of query words when an LSSS matrix is converted from a (t,n)-access policy, where t represents the minimum number of query words. Then, we have $\omega \in \mathbb{Z}_p$, so that $\sum_{w_i\in Q} \omega \cdot M_i = (1,0,\ldots,0)$. After calculating ω_i of each keyword w_i, CSP can restore the random number t based on the formula $\sum_{w_i\in Q} \omega_i \cdot \lambda_i = t$. Then, the keyword matching formula is shown in the formula (17).

$$\begin{aligned} E' &= \prod_{w_i\in Q} \left[e\left(C_w, D_j'\right) \cdot e\left(C_w', D_j\right)\right]^{\omega_{w_j}} \\ &= e(g,g)^{\alpha\beta ts_0} \end{aligned} \tag{17}$$

   C. CSP computes the mid-value E through the formula (18), and sends the mid-result $(\tilde{C}, E)$ to DS.

$$E = \frac{e\left(C', \hat{D}\right) E_R}{E'} = e(g,g)^{ads_0} \tag{18}$$

6. Decrypt$(\tilde{C}, E, d)$: After received the mid-result $(\tilde{C}, E)$, DS uses the random number d generated in TrapGen algorithm to decrypt $(\tilde{C}, E)$ for recovering the plaintext according to (19). It is noted that d is a non-public random number.

$$\frac{\tilde{C}}{E^{\frac{1}{d}}} = \frac{M \cdot e(g,g)^{\alpha s_0}}{\left[e(g,g)^{\alpha ds_0}\right]^{\frac{1}{d}}} = M \tag{19}$$

VI. Performance Analysis

In this section, we give the performance analysis, where includes the theoretical analysis and experimental simulation. It is noted that ABSE-ERM scheme is secure against the selectively chosen-keyword attack, and we omitted the proof because of space limitations.

A. Theoretical analysis

TABLE I
THE DEFINITION OF SEVERAL TIME-CONSUMING OPERATIONS IN TERMS OF COMPUTATIONAL AND STORAGE COSTS

$G_i(i=0,T)$	The computation in exponentiation or multiplication group.
$\lvert * \rvert$	The number of elements in $*$.
B	The number of keywords calculated.
S	The submitted attributed set of searchers.
W	The keyword set of ciphertext.
N	The attribute set satisfied the access policy in ciphertext.
P	The computation in bilinear pairings.

The chart will demonstrate the theoretical and actual performance of ABSE-ERM scheme through comparing with AB-HBKS scheme [36]. First of all, in order to more intuitively introduce the theoretical analysis of ABSE-ERM scheme, several time-consuming operations in terms of computational and storage cost is defined in Table I. Hereafter, for different search mechanisms, the calculation and storage costs of the algorithms of ABKS-ERM scheme and [36] are different. Thus, we mainly focus on Boolean search mechanism. Finally, we analyze the calculation and storage costs of the two schemes through Table II and Table III. as well as their causes. To be noted, there are three causes for the differences between our ABSE-ERM scheme and [36], which is the number of attributes, the number of keyword and the calculation formula. Therefore, we assume that ABSE-ERM scheme does not use 0, 1–coding theory to decrease the number of keywords. That is to say, the theoretical analysis is conducted under the premise of a certain number of keywords and attributes.

Table II illustrates the results obtained from the analysis of computational costs in Sea.Setup, TrapGen, and Search algorithm by compared with ABSE-ERM scheme and [36]. In Sea.Setup algorithm, the scheme will generate higher calculation costs, compared with [36]. As shown in Table II, It will lead higher calculation costs in ABSE-ERM scheme according to (6). The reason is that the scheme in [36] can only encrypt keyword, but it cannot provide encryption operation for plaintext. Hence, ABSE-ERM scheme introduces formula (6) to provide more complete security services. In TrapGen algorithm, compared with [36], ABSE-ERM scheme effectively reduces the calculation costs. The reason is that

TABLE II
COMPARISON OF OUR ABSE-ERM SCHEME AND SCHEME [36] IN COMPUTATIONAL COST.

Algorithm	ABSE-ERM scheme	[36]
Sea.Encrypt	$(1+2\|N\|+3\|W\|)G_0+P$	$(1+2\|N\|+3\|W\|)G_0$
TrapGen	$(2+2\|S\|+2\|M\|)G_0$	$(1+2\|S\|+3\|B\|)G_0$
Search	$(2\|N\|+2\|M\|)P+(1+\|N\|)G_T$	$(1+2\|N\|+2\|M\|)P+(\|N\|+\|B\|)G_T$

TABLE III
COMPARISON OF OUR ABSE-ERM SCHEME AND SCHEME [36] IN STORAGE COST.

Algorithm	ABSE-ERM scheme	[36]
Sea.Encrypt	$(1+2\|N\|+2\|W\|)\,\|G_0\|+\|P\|$	$(1+2\|N\|+2\|W\|)\,\|G_0\|$
TrapGen	$(2+2\|S\|+\|M\|)\,\|G_0\|$	$(1+2\|S\|+2\|B\|)\,\|G_0\|$

ABSE-ERM scheme presents the searchable LSSS matrix to implement multiple search mechanism, instead of using hybrid Boolean expression in [36]. The advantage of using searchable LSSS matrix is that the formula has been optimized in TrapGen algorithm. As illustrated in Table II, the computational efficiency of TrapGen algorithm is ideally improved by about 33.3%, compared with [36]. In Search algorithm, ABSE-ERM scheme is more efficient than [36], because LSSS matrix is more efficient than the access tree structure in transferring secret values.

Table shows the storage costs of Sea.Setup, TrapGen, and Search algorithm in ABSE-ERM scheme and [36]. Due to the linear relationship between storage costs and computational costs, ABSE-ERM scheme also has obvious advantages in terms of storage costs.

B. Experimental Simulation

In our work, the experimental simulation of ABSE-ERM scheme is implemented by JAVA. Moreover, the programs employ the Java Pairing-Based Cryptography library (JPBC) and a pairing-friendly type-A 160bit elliptic curve group. The computer configuration is shown as below.

- System: Windows 10 Professional.
- CPU: Intel Core i7 – 9700K CPU @ 3.6GHz.
- RAM: 16GB RAM.

The simulation result is the average of five times, and the unit is defined as milliseconds (ms). Besides, we assume that the number of attributes is 10. It is noted that there don't exist simulation results of storage costs in [36]. Therefore, this chart only shows the simulation comparison of computational costs between ABSE-ERM scheme and [36]. It is noted that the number of calculated keywords means the number of keywords required by CSP to complete the keyword matching process.

The histogram in Figure 2 indicates that the comparison of computational costs in TrapGen algorithm between ABSE-ERM scheme and [36]. As stated in the theoretical analysis section, the calculational costs of ABSE-ERM scheme is reduced $|W| \cdot G_0$ by the searchable LSSS matrix, compared with [36]. Therefore, from the data in Figure 2, it is obvious that the calculation efficiency of ABSE-ERM scheme is better than [36]. Besides, ABSE-ERM scheme can improve the

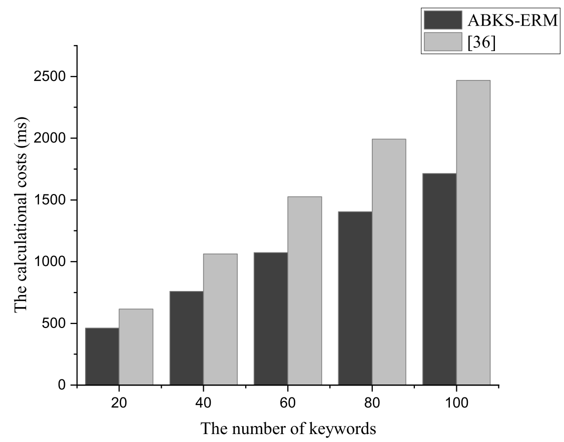

Fig. 2. The compassion of calculational costs in TrapGen algorithm.

calculation efficiency by about 33.3%, when $|W| \to \infty$. For example, the scheme is reduced by about 1000ms compared with [36], when $|W| = 100$.

Figure 3 demonstrates that the ABSE-ERM scheme without 0, 1–coding theory can be slightly improved the computational efficiency of Search algorithm, compared with [36]. However, what is interesting in this data is that there is a clear trend of increased efficiency about ABSE-ERM scheme, when the number of keywords gradually increases. For example, ABSE-ERM scheme obviously provides better calculation efficiency with 100 keywords. Among the plausible explanations for these findings is that LSSS matrix can transfer the secret value faster than hybrid Boolean expression.

Figure 4 exhibits the comparison of computational efficiency in Search algorithm between the ABSE-ERM scheme and [36], when ABSE-ERM scheme reduces the number of keywords using 0, 1–coding theory in range search mechanism. Data from this figure can be compared with the data in Figure 3, which shows the calculation costs of our scheme have always been kept at a low level in range search mechanism, even though the search scope continues to increase. The reason for this phenomenon is that DS needs to choose n

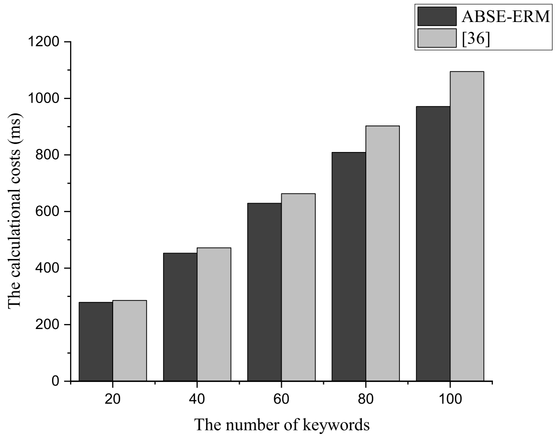

Fig. 3. The compassion of calculational costs in Search algorithm. (ABSE-ERM scheme without 0,1 – coding theory)

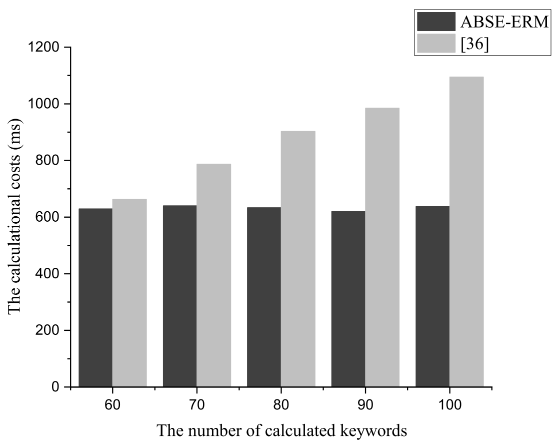

Fig. 4. The compassion of computational costs in Search algorithm. (ABSE-ERM scheme with 0, 1 – coding theory)

tags in the search range, and determine whether the keyword satisfies this range by matching whether the numeric keyword and the tag are equal. Hence, the hybrid Boolean expression needs to construct n leaf nodes associated with those tags. However, the searchable LSSS matrix with 0, 1–coding theory only needs a few rows associated with a few tags. As illustrated in Figure 4, when the number of calculated keywords is 100, the efficiency of ABSE-ERM scheme has been improved by nearly 50%.

Conclusion

The purpose of the current study is to implement an efficient and human-friendly ABSE scheme, which can provide multi-search mechanism, multi-keyword search, and so on. In this paper, we show that the searchable LSSS matrix can offer more flexible search method, and reduce the costs of this scheme with the 0, 1–coding theory. In the future, we will focus on how to implement SE scheme in new area, like blockchain area.

References

[1] L. Liu, J. Xu, Y. Huan, Z. Zou, S. Yeh and L. Zheng, “A Smart Dental Health-IoT Platform Based on Intelligent Hardware, Deep Learning, and Mobile Terminal,” in IEEE Journal of Biomedical and Health Informatics, vol. 24, no. 3, pp. 898-906, March 2020, doi: 10.1109/JBHI.2019.2919916.

[2] H. Zhang, W. Xiao and H. Jiang, “Optimal Layout of Wearable Intelligent Terminal Micro Sensor and Modeling of Elbow Movement Function Rehabilitation,” in IEEE Access, vol. 7, pp. 158881-158891, 2019, doi: 10.1109/ACCESS.2019.2950894.

[3] J. Shen, T. Zhou, D. He, Y. Zhang, X. Sun and Y. Xiang, “Block Design-Based Key Agreement for Group Data Sharing in Cloud Computing,” in IEEE Transactions on Dependable and Secure Computing, vol. 16, no. 6, pp. 996-1010, 1 Nov.-Dec. 2019, doi: 10.1109/TDSC.2017.2725953.

[4] J. Emeras, S. Varrette, V. Plugaru and P. Bouvry, “Amazon Elastic Compute Cloud (EC2) versus In-House HPC Platform: A Cost Analysis,” in IEEE Transactions on Cloud Computing, vol. 7, no. 2, pp. 456-468, 1 April-June 2019, doi: 10.1109/TCC.2016.2628371.

[5] S. J. E. Taylor et al., “Enabling Cloud-Based Computational Fluid Dynamics with a Platform-as-a-Service Solution,” in IEEE Transactions on Industrial Informatics, vol. 15, no. 1, pp. 85-94, Jan. 2019, doi: 10.1109/TII.2018.2849558.

[6] W. Li et al., “Unified Fine-Grained Access Control for Personal Health Records in Cloud Computing,” in IEEE Journal of Biomedical and Health Informatics, vol. 23, no. 3, pp. 1278-1289, May 2019, doi: 10.1109/JBHI.2018.2850304.

[7] Y. Yao, Z. Zhai, J. Liu and Z. Li, “Lattice-Based Key-Aggregate (Searchable) Encryption in Cloud Storage,” in IEEE Access, vol. 7, pp. 164544-164555, 2019, doi: 10.1109/ACCESS.2019.2952163.

[8] M. Zeng, K. Zhang, H. Qian, X. Chen, J. Chen and Y. Mu, “A Searchable Asymmetric Encryption Scheme with Support for Boolean Queries for Cloud Applications,” in The Computer Journal, vol. 62, no. 4, pp. 563-578, April 2019, doi: 10.1093/comjnl/bxy134.

[9] J. Niu, X. Li, J. Gao and Y. Han, “Blockchain-Based Anti-Key-Leakage Key Aggregation Searchable Encryption for IoT,” in IEEE Internet of Things Journal, vol. 7, no. 2, pp. 1502-1518, Feb. 2020, doi: 10.1109/JIOT.2019.2956322.

[10] S. Tahir, S. Ruj, Y. Rahulamathavan, M. Rajarajan and C. Glackin, “A New Secure and Lightweight Searchable Encryption Scheme over Encrypted Cloud Data,” in IEEE Transactions on Emerging Topics in Computing, vol. 7, no. 4, pp. 530-544, 1 Oct.-Dec. 2019, doi: 10.1109/TETC.2017.2737789.

[11] J. Ning, J. Xu, K. Liang, F. Zhang and E. Chang, “Passive Attacks Against Searchable Encryption,” in IEEE Transactions on Information Forensics and Security, vol. 14, no. 3, pp. 789-802, March 2019, doi: 10.1109/TIFS.2018.2866321.

[12] S. Wang, K. Liang, J. K. Liu, J. Chen, J. Yu and W. Xie, “Attribute-Based Data Sharing Scheme Revisited in Cloud Computing,” in IEEE Transactions on Information Forensics and Security, vol. 11, no. 8, pp. 1661-1673, 2016.

[13] S. Wang, J. Zhou, J. K. Liu, J. Yu, J. Chen and W. Xie, “An Efficient File Hierarchy Attribute-Based Encryption Scheme in Cloud Computing,” in IEEE Transactions on Information Forensics and Security, vol. 11, no. 6, pp. 1265-1277, 2016.

[14] W. Li et al., “Unified Fine-Grained Access Control for Personal Health Records in Cloud Computing,” in IEEE Journal of Biomedical and Health Informatics, vol. 23, no. 3, pp. 1278-1289, May 2019, doi: 10.1109/JBHI.2018.2850304.

[15] H. Xiong, Y. Bao, X. Nie and Y. I. Asoor, “Server-Aided Attribute-Based Signature Supporting Expressive Access Structures for Industrial Internet of Things,” in IEEE Transactions on Industrial Informatics, vol. 16, no. 2, pp. 1013-1023, Feb. 2020, doi: 10.1109/TII.2019.2921516.

[16] J. Li et al., “An Efficient Attribute-Based Encryption Scheme With Policy Update and File Update in Cloud Computing,” in IEEE Transactions on Industrial Informatics, vol. 15, no. 12, pp. 6500-6509, Dec. 2019, doi: 10.1109/TII.2019.2931156.

[17] X. Song, D. Wagner, and A. Perrig. “Practical techniques for searches on encrypted data.” Proceeding 2000 IEEE Symposium on Security and Privacy. SP 2000. IEEE, 2000, pp. 44-45.

[18] P. Golle, J. Staddon, and B. Waters. “Secure conjunctive keyword search over encrypted data.” International Conference on Applied Cryptography and Network Security. Springer, Berlin, Heidelberg, 2004, pp. 31-45.
[19] Wang, et al. “IDCrypt: a multi-user searchable symmetric encryption scheme for cloud applications.” IEEE Access 6 (2017): 2908-2921.
[20] Gao, et al. “M-SSE: an effective searchable symmetric encryption with enhanced security for mobile devices.” IEEE Access 6 (2018): 38860-38869.
[21] Hu, et al. “Forward Secure Conjunctive-Keyword Searchable Encryption.” IEEE Access 7 (2019): 35035-35048.
[22] Boneh, et al. “Public key encryption with keyword search.” International conference on the theory and applications of cryptographic techniques. Springer, Berlin, Heidelberg, 2004, PP. 506-522.
[23] R. Waters, D. Brent, et al. “Building an Encrypted and Searchable Audit Log.” in Proceedings of the Network and Distributed System Security Symposium, NDSS 2004, San Diego, California, USA, 2004, pp. 5-6.
[24] W. Sun, et al. “Verifiable privacy-preserving multi-keyword text search in the cloud supporting similarity-based ranking.” IEEE Transactions on Parallel Distributed Systems, 2013, 25(11): 3025-3035.
[25] J. Wang, et al. “Efficient verifiable fuzzy keyword search over encrypted data in cloud computing.” Comput. Sci. Inf. Syst. 10.2 (2013): 667-684.
[26] P. Xu, et al. “Generating searchable public-key ciphertexts with hidden structures for fast keyword search,” IEEE Transactions on Information Forensics and Security 10.9 (2015): 1993-2006.
[27] R. Chen, Y. Mu, G. Yang, F. Guo and X. Wang, “Dual-Server Public-Key Encryption with Keyword Search for Secure Cloud Storage,” in IEEE Transactions on Information Forensics and Security, vol. 11, no. 4, pp. 789-798, April 2016.
[28] Y. Lu, et al. “Constructing certificateless encryption with keyword search against outside and inside keyword guessing attacks.” IEEE China Communications, 2019,16(7): 156-173.
[29] R. Ahuja and S. K. Mohanty, “A Scalable Attribute-Based Access Control Scheme with Flexible Delegation cum Sharing of Access Privileges for Cloud Storage,” in IEEE Transactions on Cloud Computing, vol. 8, no. 1, pp. 32-44, 1 Jan.-March 2020, doi: 10.1109/TCC.2017.2751471.
[30] L. Cheng et al., “SCTSC: A Semicentralized Traffic Signal Control Mode with Attribute-Based Blockchain in IoVs,” in IEEE Transactions on Computational Social Systems, vol. 6, no. 6, pp. 1373-1385, Dec. 2019, doi: 10.1109/TCSS.2019.2904633.
[31] L. Zhang, Y. Cui and Y. Mu, “Improving Security and Privacy Attribute Based Data Sharing in Cloud Computing,” in IEEE Systems Journal, vol. 14, no. 1, pp. 387-397, March 2020, doi: 10.1109/JSYST.2019.2911391.
[32] Wang, W. Li, Y. Li, and X. Xu, “A ciphertext-policy attributebased encryption scheme supporting keyword search function,” in Cyberspace Safety and Security - 5th International Symposium, CSS 2013, Zhangjiajie, China, November 13-15, 2013, Proceedings, 2013, pp. 377–386.
[33] Q. Zheng, S. Xu, and G. Ateniese, “VABKS: verifiable attribute-based keyword search over outsourced encrypted data,” in Proc. IEEE Conference on Computer Communications (INFOCOM’14). IEEE, 2014, pp. 522–530.
[34] N. Wang, J. Fu, B. K. Bhargava and J. Zeng, “Efficient Retrieval Over Documents Encrypted by Attributes in Cloud Computing,” in IEEE Transactions on Information Forensics and Security, vol. 13, no. 10, pp. 2653-2667, Oct. 2018.
[35] Y. Miao, J. Ma, X. Liu, J. Weng, H. Li and H. Li, “Lightweight Fine-Grained Search Over Encrypted Data in Fog Computing,” in IEEE Transactions on Services Computing, vol. 12, no. 5, pp. 772-785, 1 Sept.-Oct. 2019, doi: 10.1109/TSC.2018.2823309.
[36] K. He, J. Guo, J. Weng, J. Weng, J. K. Liu and X. Yi, “Attribute-Based Hybrid Boolean Keyword Search over Outsourced Encrypted Data,” in IEEE Transactions on Dependable and Secure Computing, doi: 10.1109/TDSC.2018.2864186.
[37] K. Xue, J. Hong, Y. Xue, D. S. L. Wei, N. Yu and P. Hong, “CABE: A New Comparable Attribute-Based Encryption Construction with 0-Encoding and 1-Encoding,” in IEEE Transactions on Computers, vol. 66, no. 9, pp. 1491-1503, 1 Sept. 2017, doi: 10.1109/TC.2017.2693265.
[38] Y. Miao, J. Ma, X. Liu, X. Li, Z. Liu and H. Li, “Practical Attribute-Based Multi-Keyword Search Scheme in Mobile Crowdsourcing,” in IEEE Internet of Things Journal, vol. 5, no. 4, pp. 3008-3018, Aug. 2018, doi: 10.1109/JIOT.2017.2779124.
[39] J. Bethencourt, A. Sahai, and B. Waters, “Ciphertext-policy attribute-based encryption,” in Proc. IEEE Symp. Security Privacy (SP’07), 2007, pp. 321–334.
[40] Beimel, A.: Secure schemes for secret sharing and key distribution. Ph.D. dissertation. Israel Institute of Technology, Technion, Haifa, Israel, 1996.
[41] Beimel A, Farràs O, Mintz Y, et al. “Linear Secret-Sharing Schemes for Forbidden Graph Access Structures,” in Theory of Cryptography Conference (TCC 2017). Springer, Cham, 2017:394-423.
[42] Shamir A. How to Share a Secret[J]. Comm Acm, 1979: 159-168.